

Secondary Raynaud Phenomenon

Connective Tissue Disease

- **Scleroderma:** 80% to 90% of patients manifest Raynaud phenomenon or persistent vasospasm. It is the presenting symptom in about one-third of cases and may be the sole manifestation for years.
- **Systemic lupus erythematosus:** Raynaud phenomenon occurs in approximately one-third of patients.
- **Dermatomyositis and polymyositis:** Also associated with Raynaud phenomenon in about one-third of cases.
- **Undifferentiated connective tissue disease**
- **Systemic vasculitis**
- **Sjögren syndrome**
- **Eosinophilic fasciitis**

Arteriograms in connective tissue diseases often reveal obstructions in digital arteries, and sometimes in ulnar or radial arteries.

Occupational and Environmental Exposure

- **Vibration-induced injury:** Common in workers using vibratory tools (e.g., air hammers, chain saws, rivet guns). Prevalence can reach up to 90% in high-risk occupations such as logging or mining.
- **Cold exposure:** Seen in occupations involving prolonged cold exposure (e.g., butchers, ice cream workers, fish packers).
- **Post-traumatic injury:** Includes hypothenar hammer syndrome and crutch-induced vascular injury.
- **Vinyl chloride disease**
- **Cold injury**

Neurologic Disorders

Any neurologic condition causing permanent limb disuse may lead to sympathetic nervous system dysfunction, resulting in: - Persistent vasospasm - Coldness, pallor, or cyanosis - Ulcerations - Raynaud phenomenon

Conditions include: - **Carpal tunnel syndrome**: Raynaud phenomenon often affects the first, index, and middle fingers (median nerve distribution). - **Reflex sympathetic dystrophy**: Thermoregulatory abnormalities are a prominent feature. - **Hemiplegia** - **Polio** - **Multiple sclerosis** - **Syringomyelia** - Nerve root compression or entrapment

Neurovascular compression at the **thoracic outlet** (e.g., due to cervical ribs, scalenus anticus abnormalities, bony deformities, or shoulder syndromes like costoclavicular or hyperabduction syndrome) can also cause Raynaud phenomenon.

Drugs and Toxins

Medications and substances associated with Raynaud phenomenon include: - **β -Adrenergic blockers** (e.g., propranolol) – commonly implicated in cardiovascular and migraine treatment - **Ergotamines** and **methysergide** – used for migraines; can induce vasospasm - **Oral contraceptives** - **Clonidine** - **Bromocriptine** - **Cyclosporine** - **Amphetamines** - **Fluoxetine** - **Interferon- α** - **Bleomycin** and **vinblastine**

Obstructive Arterial Disease

- **Atherosclerosis**
- **Thromboangiitis obliterans (Buerger disease)**
- **Thromboembolism**
- **Thoracic outlet syndrome**

Hyperviscosity Disorders

- **Cryoproteins**
- **Cold agglutinins**
- **Macroglobulins** (e.g., Waldenström macroglobulinemia)
- **Polycythemia**
- **Thrombocytosis**

Miscellaneous

- **Hypothyroidism**
- **Infections:** Bacterial endocarditis, Lyme disease, viral hepatitis
- **Neoplasms**
- **Primary pulmonary hypertension**
- **Arteriovenous fistula**
- **Intra-arterial injections**